

JIXIAN LIU

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APPOINTMENTS

Graduate Research Assistant

Aug. 2025 - Present

NetD-Lab, Data Science and AI Institute, Advisor: Enrique Mallada
Johns Hopkins University

RESEARCH INTERESTS

Robotics, Robot Learning, Safety-Critical Control, Reinforcement Learning.

EDUCATION

Johns Hopkins University

Aug. 2024 - Present

Ph.D. Electrical and Computer Engineering

Xi'an Jiaotong University

Sept. 2020 - Jul. 2024

B.S. Physics

HONORS & AWARDS

- Amazon AI PhD Fellowship (\$155,000) 2026
- ACC Travel Grant 2026
- Mathematical Institute for Data Science Fellowship (\$25,000) 2026
- Hong Kong PhD Fellowship (Top Award for HK PhD Applicants) 2024
- Dean's Fellow Award, Boston University (Top Award for BU PhD Applicants) 2024
- MITACS Globalink Research Internship Scholarship (C\$7000) 2023
- Top-10 Undergraduate Students of the Year (Top Individual Award in XJTU) 2023
- National Scholarship (Top 1%) 2022, 2023
- Finalist Winner, Mathematical Contest in Modeling (Top 2% around the World) 2022
- First Place in Shaanxi, The Chinese Mathematics Competitions 2021
- Presidential Scholarship of Xi'an Jiaotong University 2021

RESEARCH PUBLICATIONS

Conferences & Workshops

1. **Jixian Liu**, Ihab Tabbara, Hussein Sibai and Enrique Mallada, "On the Limits of Sampling-Based Reachability: Geometry, Dynamics, and Sample Complexity", 10th Annual Conference on Robot Learning, 2026.
2. Zhuo Ouyang, **Jixian Liu**, and Enrique Mallada, "Symplectic Inductive Bias for Data-Driven Target Reachability in Hamiltonian Systems", 65th IEEE Conference on Decision and Control, 2026.
3. **Jixian Liu**, and Enrique Mallada, "Safety-Critical Control via Recurrent Tracking Functions", 2026 American Control Conference, 2026.
4. **Jixian Liu**, and Enrique Mallada, "Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification", 64th IEEE Conference on Decision and Control, 2025.
5. **Jixian Liu**, Tao Xu, Jianping He, and Chongrong Fang, "Smart Predict-then-Optimize Method with Dependent Data: Risk Bounds and Calibration of Autoregression", arXiv preprint arXiv:2411.12653, 2024.

Journals

1. Xueyuan Cui, **Jixian Liu**, and Yao Zhou, “Enabling Net Zero Energy Buildings With Shared Storage: A Cyber-Physical Perspective”, IEEE Internet of Things Journal, 2023.

ACADEMIC SERVICES

Reviewer of Journals

- IET Renewable Power Generation Journal
- IEEE Control Systems Letters

Reviewer of Conferences & Workshops

- American Control Conference 2026, 2027
- IEEE Conference on Decision and Control 2026
- China Automation Congress 2024

TEACHING

Teaching Assistant

- Control System Design, Johns Hopkins University Fall 2025
- Foundation of Reinforcement Learning (Online), Johns Hopkins University Fall 2025, Fall 2026

ADVISING AND MENTORING

Mentoring

Zhuo Ouyang Undergrad Student, Peking University
College of Engineering Next position: CS Ph.D. at UIUC (Advisors: Tong Zhang and Nan Jiang)

PRESENTATIONS

Talks

- May 2026, “Safety-Critical Control via Recurrent Tracking Functions”, American Control Conference, New Orleans, LA, USA
- Dec. 2025, “Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification”, 64th IEEE Conference on Decision and Control, Rio de Janeiro, RJ, Brazil
- Oct. 2025, “Recurrence is All You Need: From Parametric to Nonparametric Safety”, Institute for Assured Autonomy Graduate Student Lightning Talks, Baltimore, MD, USA

Poster

- May 2025, “Recurrent Control Barrier Functions: A Path Towards Nonparametric Safety Verification”, 2nd Northeast Systems and Control Workshop, NYC, USA

REFERENCES

Enrique Mallada Ph.D. Advisor
Associate Professor, Department of Electrical and Computer Engineering
Johns Hopkins University, Baltimore, MD

Jianping He
Professor, School of Automation and Intelligent Sensing
Shanghai Jiao Tong University, Shanghai, SH

Xiaoming Duan
Associate Professor, School of Automation and Intelligent Sensing
Shanghai Jiao Tong University, Shanghai, SH